

# Math GRE Prep Day 2

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# Outline

- 1 Calculus Problems
  - Standard problems
  - Somewhat trickier problems

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# Problem 1

Suppose that  $f(x)$  is twice differentiable with  $f(2) = 10$  and  $f'(x) = x^2 f(x)$ . Find  $f''(2)$ .

# Problem 2

Find

$$\lim_{x \rightarrow 0} \frac{\sin(3x) \sin(2x)}{x \tan(6x)}.$$

# Problem 3

How many tangent lines to the curve  $y = x^2$  pass through the point  $(3, -7)$  and what are their slopes?

# Problem 4

The tangent line to the curve  $y = f(x)$  at  $x = 2$  is  $y - 8 = 2(x - 2)$ . Calculate

$$\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2}$$

# Problem 5

Suppose that  $f(x) = |x - 1| - |x + 1|$ . Compute  $\int_{-3}^3 (f(x - 1) - f(x + 1) + 1) dx$



# Problem 6

Find an integral expression which is equal to

$$\lim_{n \rightarrow \infty} \frac{3}{n^2} \left( \sqrt{4n^2 + 9 \times 1^2} + \sqrt{4n^2 + 9 \times 2^2} + \cdots + \sqrt{4n^2 + 9 \times n^2} \right)$$

# Problem 7

A region is bounded by two concentric circles, where the smaller circle has radius  $r$  and the bigger circle has radius  $R$ . If  $r$  decreases at 1 inch per second and  $R$  increases at 2 inches per second, how fast is the area changing at the instant when  $r = 3$  inches and  $R = 4$  inches?

# Problem 8

Let  $f(x) = x^{2024}$ . Evaluate

$$f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \cdots + \frac{f^{(2024)}(1)}{2024!}$$

# Problem 9

Find the smallest positive integer  $m$  such that  $m^{1/m} \geq n^{1/n}$  for all positive integers  $n \geq m$ .

# Problem 10

Find all values of  $x$  such that

$$\lim_{n \rightarrow \infty} (x + x^3 + x^5 + \cdots + x^{2n-1}) = \frac{3}{8}.$$